Minghan Lyu

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EDUCATION

Shandong University (SDU), Jinan, China Sep2023-Present
Bachelor of Engineering in Software Engineering (Emerging Engineering Honors Program in Artificial Intelligence), in progress at the School of Software. The program is offered under the Joint SDU-NTU Centre for Artificial Intelligence Research (C-FAIR).
Related courses: Numerical Computing (92), Information Retrieval (92), Linear Algebra (91), Computer Organization and Architecture (90), Introduction to the Artificial Intelligence Major (95), Social Networks and Recommender Systems (90)

RESEARCH EXPERIENCE

Machine Learning for Protein Modeling Jun 2026 - Sep 2026
Advisor: Sorin Istrail (Department of Computer Science, Brown University; Professor)
- Led a three-member course project on CGnet/CGSchNet transferability; developed the CGSchNet analysis and integrated the final report and presentation.
- Independently audited published data for 27 ubiquitin mutations and two protein fluctuation profiles using bootstrap, sensitivity, and structure-localization analyses.
- Built a separate ATLAS benchmark from 200 protein chains with three MD replicas each and an ECOD-component-disjoint 132/38/30 train/validation/test split.
- Combined contact-persistence prediction with weighted Gaussian network models; achieved test persistence MAE 0.0856 and Spearman 0.8878.
- Improved mean test RMSF Spearman from 0.7170 to 0.7229 over uniform GNM; validated the modest gain with component-level resampling and ablations.

Policy-Guided Mango Shape Annealing Apr 2026 - Aug 2026
Advisor: John H. Reif (Trinity College of Arts and Sciences, Duke University; Professor)
- Reframed Mango's stochastic shape-annealing search as masked, constraint-aware proposal guidance for wireframe DNA origami design.
- Engineered a hierarchical grammar-action interface over rule, site, and parameter choices with legal action masking and failure diagnostics.
- Built reproducible B1/B2/B3 benchmark pipelines with unified logging, constraint-failure analysis, candidate audits, and baseline comparisons.
- Demonstrated PG-Mango gains over Mango ShapeAnneal and Random-Feasible on 5/5 B1-family benchmarks while preserving manufacturability constraints.
- I have open-sourced this project on GitHub.

Eukaryotic Gene Structure Prediction with HMM Oct 2025 - Jan 2026
Advisor: Jun Wang (School of Software, Shandong University; Professor)
- Modeled gene boundaries as repeat-sensitive labeling with auditable, aligned FASTA/GFF3 samples and splits.
- Implemented a multi-state HMM with CDS phase modeling and standardized GFF3 decoding for gene prediction.
- Used nucleotide/exon Sn-Sp for iterative, targeted refinements; open-sourced the gene prediction code on GitHub.

Traffic Forecasting on METR-LA Sep 2025 - Dec 2025
Advisor: Yongshun Gong (School of Software, Shandong University; Professor)
- Built a multi-step forecasting pipeline from cleaning and windowing to masked evaluation with missing data.
- Captured non-stationary spatiotemporal dependencies with graph priors and multi-branch sequence learning.
- Reduced test MAE/RMSE/MAPE to 3.57/7.37/10.17% on METR-LA; open-sourced the project code on GitHub.

News Recommendation with NPA Jan 2025 - Jun 2025
Advisor: Li Lian (School of Software, Shandong University; Associate Professor)
- Reproduced NPA on MIND as a reliable baseline, building end-to-end impression and user-history pipelines.
- Used controlled negative sampling and Transformer-style encoders to learn richer semantic representations.
- Improved offline ranking metrics (AUC/MRR/nDCG@K) through stable training and systematic model tuning.

LLM-based Paper Screening Nov 2024 - Dec 2024
Advisor: Yuqing Sun (School of Software, Shandong University; Professor)
- Structured title/abstract prompts for systematic reviews around inclusion/exclusion criteria and project constraints.
- Used LLM classification with reproducible logs and lightweight checks for faster screening and consistent quality.

PROJECT EXPERIENCE

Yantu Lingban: Agentic Exam-Preparation Companion System Mar 2026 - Jun 2026
Advisor: Ning Liu (School of Software, Shandong University; Assistant Researcher)
- Constructed an agentic Windows desktop companion system for postgraduate-exam preparation, covering study planning, subject question answering, mistake review, proactive care, and emotion/diet management.
- Designed a multi-agent workflow with a central orchestrator, specialized study/life/psychology agents, and a shared user-state center for context-aware decisions.
- Integrated rule-driven triggers for active interventions, spaced mistake review, and adaptive plan adjustment based on learning progress, pending review tasks, and user status.
- Connected Electron desktop functions, FastAPI services, WebSocket streaming, OCR-based screenshot question answering, and report generation into end-to-end study loops.
- I have open-sourced this project on Gitee.

Multimodal Understanding for Product Live-stream Sep 2025 - Dec 2025
Advisor: Lei Meng (School of Software, Shandong University; Professor)
- Aligned live-stream keyframes, speech text, and product metadata in a reproducible pipeline for product retrieval.
- Mined hard negatives via embedding similarity to sharpen fine-grained contrastive learning across modalities.
- Reduced noise via ROI localization and stabilized fusion with late ensembling; open-sourced the code on GitHub.

Brand-Geographic Grid Prediction Mar 2025 - Jul 2025
Advisor: Meng Chen (School of Software, Shandong University; Associate Professor)
- Cast site selection as next-grid prediction on sparse, weakly timestamped trajectories; built grid-aligned sequences.
- Used density-aware ordering and sliding-window supervision; fused trajectory and geographic/context features.
- Achieved test Acc@1/5/10 of 0.270/0.567/0.724 and MRR of 0.450 with a multimodal next-grid predictor.

INTERNSHIP

Invoice Reimbursement Review System | Advisor: Siwei Chen Jul 2026
Intern | Research and Practical Teaching Center, Chinese Aeronautical Establishment
- Developed browser-based student submission and local Python teacher review workflows for reimbursement materials.
- Integrated PDF/Word parsing, OCR, and optional vision-language models for invoice extraction and supporting-document checks.
- Implemented invoice-level reconciliation, manual review, Excel/PDF exports, and portable Windows packaging.
- Assisted Coze-based AI programming labs for Khalifa University exchange students.

PUBLICATION

Lyu, M. (2026). Predicting Molecular Dynamics Contact Persistence from Static Structures for Weighted Gaussian Network Models. Theoretical and Natural Science, 190, 142-154.
ICBioMed 2026 symposium: Biomedical Engineering and Smart Medicine (ICBM_HZ 2026).
DOI: 10.54254/2753-8818/2026.HZ37271

AWARDS AND HONORS

- Mathematical Modeling National Contest, Problem C, Provincial First Prize (8%) 2 times in Year 2024 and 2025
- Innovation Research Specialty Scholarship 2024-2025 Academic Year
- 2026 COMAP Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling (MCM/ICM), Meritorious Winner (M Award) (6.67%)

SKILLS AND OTHERS

Programming language: C/C++, Python, Java, MySQL, MATLAB, LaTeX, R
Framework & Tools: PyTorch, GitHub, TensorFlow, Java IDE, Spring Web, Thymeleaf, Hugging Face, ROS, Ubuntu, Anaconda, BLAST, MEME Suite, RStudio, Git